Spiral Similarity

Facts.

- Let A, B, C, D be distinct points on the plane such that AC and BD are not parallel, AB and CD are not parallel. Suppose the lines AC and BD intersect at P. Then the second intersection point O of (PAB) and (PCD) is the unique centre of spiral similarity that sends A to C and B to D respectively. (This means $\triangle OAB \sim \triangle OCD$ and $\triangle OAC \sim \triangle OBD$.)
- Suppose two circles Γ_1 and Γ_2 intersect at points A and B. A line passing through A meets Γ_1 and Γ_2 again at P and Q respectively. Another line passing through A meets Γ_1 and Γ_2 again at X and Y respectively. Then $\triangle BPQ \sim \triangle BXY$ and $\triangle BPX \sim \triangle BQY$.
- Let $ABCD$ be a convex quadrilateral. The lines AB and CD meet at E. The lines BC and AD meet at F. Then $(EAD), (EBC), (FAB), (FCD)$ pass through a point, called the ***Miquel point*** of $ABCD$.

Basic Problems.

Problem 1. In $\triangle ABC$, let D be a point on the internal angle bisector of $\angle BAC$ such that A and D lie on different sides of BC. The circumcircle of $\triangle ABD$ meets the extension of the side AC again at E where C lies between A and E. The circumcircle of $\triangle ACD$ meets the extension of the side AB again at F where B lies between A and F. Prove that $BF = CE$.

Problem 2. Let $ABCD$ be a convex quadrilateral. The diagonals AC and BD intersect at E. The sidelines AB and CD intersect at F. Let M and N be the midpoints of AB and CD respectively. The circumcircles of $\triangle ABE$ and $\triangle CDE$ meet at E and P. Prove that F, M, P, N are concyclic.

Problem 3. Let $ABCD$ be a cyclic quadrilateral. Let E and F be the feet of projections of D onto AC and BC respectively. Let M and N be the midpoints of AB and EF. Prove that $DN \perp MN$.

Problem 4. Let $ABCD$ be a convex quadrilateral such that AB and CD are not parallel. Let E and F be points on the sides AD and BC respectively such that $AE : ED = BF : FC$. Prove that $ABFE$ and $CDEF$ share the same Miquel point.

Problem 5. Let MN be a line parallel to the side BC of a triangle ABC, with M on the side AB and N on the side AC. The lines BN and CM meet at point P. The circumcircles of triangles BMP and CNP meet at two distinct points P and Q. Prove that $\angle BAQ = \angle CAP$.

Other Problems.

Problem 6. The diagonals AC and BD of a convex quadrilateral meet at P. Let O_1 and O_2 be the circumcentres of $\triangle APD$ and $\triangle BPC$ respectively. Let M, N and O be the midpoints of BD, AC and O_1O_2 respectively. Prove that O is the circumcentre of $\triangle PMN$.

Problem 7. Given $\triangle ABC$, let A' be the point such that $\angle ABA' = \angle ACA' = 90°$. Let D and E be points on the sides AB and AC respectively such that $ADA'E$ is a parallelogram. The lines DE and BC meet at F. The circumcircles of $\triangle ABC$ and $\triangle ADE$ meet again at X. Prove that FX is tangent to the circumcircle of $\triangle ABC$.

Problem 8. Let C be a point on the segment AE. Two equilateral triangles ABC and CDE are constructed on the same side of AE. Let O_1 and O_2 be the centres of $\triangle ABC$ and $\triangle CDE$ respectively. The circumcircles of $\triangle ABC$ and $\triangle CDE$ meet again at F. The segments O_1O_2 and AD meet at P. Prove that $AP = BF$.

Problem 9. Let Γ_1 and Γ_2 be two circles of unequal radii, with centres O_1 and O_2 respectively, intersecting in two distinct points A and B. Assume that the centre of each circle is outside the other circle. The tangent to Γ_1 at B intersects Γ_2 again in C, different from B; the tangent to Γ_2 at B intersects Γ_1 again at D, different from B. The bisectors of $\angle DAB$ and $\angle CAB$ meet Γ_1 and Γ_2 again in X and Y, respectively. Let P and Q be the circumcentres of triangles ACD and XAY, respectively. Prove that PQ is the perpendicular bisector of the line segment O_1O_2.

Problem 10. Triangle ABC with it's circumcircle Γ is given. Points D and E are chosen on segment BC such that $\angle BAD = \angle CAE$. The circle ω is tangent to AD at A with it's circumcenter lies on Γ. Reflection of A through BC is A'. If the line $A'E$ meets ω at L and K, then prove either BL and CK or BK and CL meet on Γ.